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Standard compliance bounds origin–destination identifiability in GBFS bike-sharing feeds: a cost-learning and sampling-horizon analysis

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Abstract

Reconstructing origin–destination (OD) flows from public bike-sharing feeds is split between two regimes: macroscopic inference from station stock counts, which is underdetermined, and deterministic tracking of per-vehicle identifiers, which is exposed to temporal aliasing and to the privacy-driven rotation of identifiers recommended by the GBFS standard. We recast the problem as a hybrid in which a partial, identifier-tracked micro-sample supplies the deterrence (cost) structure of an entropic optimal-transport / gravity model whose margins are pinned by the station counts. Two results follow. First, the transport cost is identifiable only modulo additive station effects; the bias induced by incomplete observation equals the *non-separable* component of the log-selection probability, so station-emptiness censoring cancels exactly, whereas 60-second polling—whose capture probability is a function of travel time, the very quantity being estimated—does not. Second, the estimator yields a *sampling-horizon law*: the continuous-collection time needed to reconstruct OD to target accuracy δ scales as δ^{-4} and inversely with the identifier-persistence rate q, implying a standard-compliance threshold below which operational OD monitoring is infeasible. An audit of **[To confirm:** N**]** French free-floating networks instantiates the bound and exhibits a sharp bimodal split between standard-compliant (identifier-rotating, untrackable) and non-compliant (persistent, trackable) feeds.

1 Introduction

Origin–destination (OD) matrices are the primitive of every operational and planning question in shared micromobility: fleet rebalancing, station siting, demand forecasting, and the equity audits that increasingly condition public funding all consume an OD matrix as input. Yet the data that operators expose publicly was never designed to carry it. The General Bikeshare Feed Specification (GBFS), the de-facto open standard adopted by well over a thousand systems worldwide, is a *real-time availability* interface: it advertises which vehicles are where, right now, and explicitly does not record what happened between two states. OD must therefore be *reconstructed*, and the literature is cleaved along the only two routes the data permits.

The first route is macroscopic: treat the per-station stock changes as link or node counts and invert. This is the classical OD-from-counts problem, and it is mathematically mature—but it is also *underdetermined*: many matrices reproduce the same observed marginals, so the interior of the OD matrix is fixed by a prior, not by the data. The second route is microscopic: when a feed exposes a persistent per-vehicle identifier, individual vehicles can be tracked between snapshots and their trips read off directly. This route is exact in principle, but fragile in practice for two reasons that the literature treats as nuisances rather than first-class objects. It is degraded by *temporal aliasing*, because polling at a finite cadence misses short or fast trips; and, more fundamentally, it is *deliberately defeated by the standard itself*, whose privacy guidance recommends rotating the vehicle identifier after every trip precisely to prevent the trip reconstruction the method relies on.

This places the analyst in a paradox that, to our knowledge, has never been made quantitative: the very property that makes a feed useful for OD reconstruction (identifier persistence) is the property the standard asks operators to remove. We argue that the feasibility of OD reconstruction is therefore not a modelling choice but a *function of the feed's compliance with the standard*, and we make that dependence precise.

Contributions. We (i) recast OD reconstruction as a hybrid in which an identifier-tracked micro-sample supplies the cost structure of an entropic optimal-transport (equivalently, maximum-entropy gravity) model whose margins are pinned by the station counts (Section 3); (ii) prove that the learned cost is identifiable only modulo additive station effects, and decompose the observation bias into the *non-separable* part of the log-selection probability, which shows that station-emptiness censoring cancels while 60-second aliasing does not (Section 4); (iii) derive a sampling-horizon law $T^\star \propto \delta^{-4} q^{-1}$ and the standard-compliance threshold it implies, and identify cross-network pooling as the only lever that keeps the horizon finite (Section 5); and (iv) instantiate the bound through a feed-by-feed audit of **[To confirm:** N**]** French networks, whose identifier-persistence rates split sharply into compliant and non-compliant regimes (Section 6). Section 7 translates the threshold into a concrete recommendation for the standard's maintainers. Because the novelty of the contribution rests on *not* re-deriving the classical estimators, we first delimit what the existing two literatures already settle.

2 Background and related work

Counts give an underdetermined, prior-driven OD. Recovering an OD matrix from link or node counts is a 45-year-old, well-posed but underdetermined problem. Van Zuylen and Willumsen (1980) introduced the maximum-entropy and minimum-information estimators; Cascetta (1984) incorporated survey priors through generalized least squares; Spiess (1987) gave the Poisson maximum-likelihood form; and Cascetta and Nguyen (1988) unified the family. A practical point that frames our own method is that the workhorse estimators are *the same object*: the iterative proportional fitting used to balance a matrix to target margins is exactly the Sinkhorn iteration for entropic optimal transport (Sinkhorn, 1964; Cuturi, 2013; Peyré and Cuturi, 2019), which is in turn the maximum-entropy gravity model of Wilson (1967). Modern work sharpens the conditioning of the inverse (Ma and Qian, 2023) but does not change the verdict: from counts alone, pair-level OD is weakly identified.

Identifier tracking is exact but treats its biases as nuisances. When a feed exposes a persistent identifier, trips are reconstructed by tracking position jumps between snapshots; Xu et al. (2022) give the reference algorithm family, validated on one week of feeds from six vendors in a single city. This and adjacent studies catalogue the biases—rebalancing contamination, GPS jitter, and aliasing under coarse polling—but treat them as cleanup steps, and are almost universally single-city, single-window, and silent on censored demand (Rudloff and Lackner, 2014; Negahban, 2019; Goh et al., 2019) and on the standard's identifier-rotation guidance.

The gap. No prior work models the impact of a *standard specification*—concretely, the identifier-rotation (persistence) rate q—on the mathematical solvability of the downstream OD model. The closest cross-domain analogue, network tomography (Vardi, 1996), draws its identifying power from a known routing matrix that the station-count setting simply does not possess, so it collapses to bare margin balancing. We close this gap by making compliance a parameter of the estimator and bounding identifiability and collection cost as functions of it. We now fix notation and the structural assumption that makes this possible.

3 Problem formulation and data regimes

Let stations be indexed $i, j \in \{1, \ldots, N\}$ and let $P^\star \in \mathbb{R}^{N\times N}_{\geq 0}$ be the true OD coupling over a fixed period, with row margins $r_i^\star = \sum_j P_{ij}^\star$ (outflow) and column margins $c_j^\star = \sum_i P_{ij}^\star$ (inflow). Throughout we adopt the Gibbs (gravity / entropic optimal-transport) structural assumption

$$P_{ij}^\star = u_i^\star \, v_j^\star \, e^{-c_{ij}^\star/\varepsilon}, \tag{1}$$

with potentials $u^\star, v^\star$, ground cost $c^\star$, and temperature $\varepsilon > 0$. As recalled in Section 2, this is simultaneously the maximum-entropy trip-distribution model and the fixed point of Sinkhorn balancing, so (1) is not an additional modelling commitment beyond the estimators the field already uses—only an explicit one.

Two observation channels constrain different parts of (1).

Macro channel: censored margins. Station-status counts provide noisy, right-censored margins $\hat{r}_i = \rho_i r_i^\star$ with availability fraction $\rho_i \in (0, 1]$, reflecting demand lost at empty or full stations (Rudloff and Lackner, 2014; Negahban, 2019). The macro channel thus pins (a degraded version of) the margins but says nothing about the interior.

Micro channel: a selection-tilted sample. A sub-sample of identifier-tracked trips is drawn not from $P^\star$ but from the *selection-tilted* law $\tilde{P}_{ij} \propto P_{ij}^\star S_{ij}$, where the selection probability $S_{ij} \in [0, 1]$ aggregates identifier persistence q, non-rebalancing $(1 - \beta)$, and capture under polling interval Δ (anti-aliasing) κ. The micro channel thus carries information about the interior—the cost—but through a biased lens. The remainder of the paper asks what that lens lets us recover, and at what cost in collection time.

4 Cost identifiability and the structure of observation bias

4.1 The transport cost is identifiable only modulo station effects

The Gibbs form (1) is invariant under $c_{ij}^\star \mapsto c_{ij}^\star + f_i + g_j$, since any additive separable term is absorbed into the potentials. The cost is therefore identifiable only modulo the separable subspace $\mathcal{N} = \{(f_i + g_j)_{ij}\}$ ($\dim \mathcal{N} = 2N - 1$); the identifiable object is the interaction part $\Pi_{\mathcal{N}^\perp} c^\star$, equivalently the second-order cross-differences $c_{ij} - c_{il} - c_{kj} + c_{kl}$.

Theorem 1 (Cost identifiability)**.** *Under (1), $\Pi_{\mathcal{N}^\perp} c^\star$ is identifiable from a coupling, and no separable component of $c^\star$ is.* ***[To confirm: State regularity conditions; the proof, a matching/ground-metric argument in the manner of Galichon and Salanié (2022); Dupuy and Galichon (2014); Cuturi and Avis (2014), is given in SI Appendix A.]***

This is not a technicality but a design rule for the cost model. A low-rank parametric cost $c_{ij} = \phi(x_i, x_j; \theta)$ may only use features whose information survives projection onto $\mathcal{N}^\perp$.

Definition 2 (Admissible cost feature)**.** A candidate feature $\psi_{ij} = \phi^{(k)}(x_i, x_j)$ is *admissible* iff it survives two-way (origin/destination) centring, $\tilde{\psi}_{ij} = \psi_{ij} - \bar{\psi}_{i\cdot} - \bar{\psi}_{\cdot j} + \bar{\psi}_{\cdot\cdot} \neq 0$. Monadic features ($\psi_{ij} = a_i$ or b_j) are inadmissible: they are absorbed into the potentials and belong in the margins, not the cost.

Admissible, mutually non-collinear dyadic features—routed cycling distance, *directed* elevation gain, circuity, protected-lane fraction, and river/rail barrier crossings—are what give the projected information matrix a well-conditioned smallest eigenvalue $\lambda_{\min}$, which Section 5 shows is what keeps the collection horizon finite.

4.2 Observation bias equals the non-separable log-selection

Because the micro-sample is drawn from $\tilde{P} \propto P^\star \odot S$ rather than $P^\star$, fitting (1) to it recovers the cost of $\tilde{P}$, namely $\tilde{c} = c^\star - \varepsilon \log S$.

Proposition 3 (Bias = non-separable log-selection)**.** *The estimated identifiable cost satisfies*

$$\Pi_{\mathcal{N}^\perp} \hat{c} \xrightarrow{n \to \infty} \Pi_{\mathcal{N}^\perp} c^\star \; - \; \varepsilon \, \Pi_{\mathcal{N}^\perp} \log S. \tag{2}$$

In particular, if the selection is separable, $S_{ij} = a_i b_j$, then $\log S \in \mathcal{N}$ and the bias vanishes. ***[To confirm: Proof in SI Appendix A.]***

4.3 Station emptiness cancels; 60-second aliasing does not

Proposition 3 converts a question about data quality into a question about the *separability* of the selection mechanism, and the two dominant defects fall on opposite sides of that line.

Corollary 4 (Which censoring hurts)**.** *Station-emptiness censoring acts at the origin/destination level ($S_{ij} = a_i b_j$) and therefore* cancels *in the identifiable cost. Aliasing under polling interval Δ has a capture probability that depends on the trip duration t_{ij}, hence on $c_{ij}^\star$ itself; it is* non-separable *and correlated with the estimand, so it does not cancel and systematically attenuates the cost–distance relationship. Residual operator rebalancing, concentrated on specific corridors, is likewise non-separable.*

The practical reading is uncomfortable: the defect one would expect to be fatal (empty stations) is harmless to the cost, while the seemingly benign engineering choice of a polling cadence is the true adversary, because it attacks exactly the signal being estimated. Since the aliasing bias can only be beaten by sampling more often and for longer, the natural next question is precisely *how long* one must collect—which leads us to formulate the sampling-horizon law.

5 The sampling-horizon law

Let Λ denote the trip-generation rate (trips per day). After T days of polling at interval Δ, the number of clean, attributable observations is

$$n_{\text{eff}} = \Lambda\, T\, \kappa(\Delta)\, q\, (1-\beta)\, (1-\gamma), \tag{3}$$

where γ is the GPS-jitter / collision loss. Propagating the cost-learning error through the entropic-OT forward map—whose sensitivity is $O(1/\varepsilon)$ while its entropic bias is $O(\varepsilon)$, so that the optimal balance yields an $n^{-1/4}$ rate on the OD error—turns a target accuracy into a required effective sample $n_{\text{eff}} \gtrsim \dim\theta/(\lambda_{\min}\, \delta_{\text{OD}}^4)$, and hence a required collection time.

Theorem 5 (Sampling-horizon law)**.** *To reconstruct the OD coupling to relative accuracy δ_{OD} under the estimator of Sections 4.1–4.3,*

$$T^\star \;=\; \frac{\dim\theta}{\lambda_{\min}\, \delta_{\text{OD}}^4\; \Lambda\, \kappa(\Delta)\, q\, (1-\beta)(1-\gamma)}, \tag{4}$$

so $T^\star \propto \delta_{\text{OD}}^{-4}\, q^{-1}\, \Lambda^{-1}\, \lambda_{\min}^{-1}$. ***[To confirm: Forward-stability lemma and proof in SI Appendix B.]***

The fourth-power dependence on accuracy is the signature of the entropic estimator: tightening the OD target from 20% to 10% multiplies the required horizon by $2^4 = 16$. The inverse dependence on q is the policy-relevant term.

Corollary 6 (Standard-compliance threshold)**.** *For any operational horizon $T_{\max}$ and accuracy δ_{OD}, OD is reconstructible only on feeds with identifier persistence $q \geq q_{\min} := \dim\theta/(\lambda_{\min}\delta_{\text{OD}}^4\Lambda\kappa(1-\beta)(1-\gamma)\,T_{\max})$. Below $q_{\min}$ the horizon exceeds $T_{\max}$ and operational monitoring is infeasible.*

5.1 Partial pooling across networks is the only lever on $T^\star$

For a single small network, (4) can run to years (Section 6). The only term the analyst controls is $\dim\theta$. Modelling the cost hierarchically—a common deterrence law θ shared across the N networks with city-level random effects—divides the per-city parameter budget by the number of poolable cities, and is therefore the mechanism that brings $T^\star$ back from "years" to "weeks". The multi-system panel is thus *statistically load-bearing*, not merely a comparative convenience; a fixed ε imposed for cross-city comparability additionally caps the achievable accuracy at $\delta_{\text{OD}} \gtrsim \varepsilon$. With the law in hand, we turn to measuring its constants on real feeds.

6 Auditing the French micromobility feeds

6.1 Protocol: measuring the five governing constants

The five constants $(q, \kappa, \beta, \gamma, \Lambda)$ are feed-specific and *measured*, not assumed. A polling daemon collects `free_bike_status`/`vehicle_status` snapshots at $\Delta = 60$ s across the French free-floating systems that expose a vehicle feed, writing one record per vehicle per poll (identifier, position, status flags). Two reproducible scripts realise the estimators: `experiments/d01_pilot.py` computes the horizon (4) from a set of constants, and `experiments/d02_persistence_audit.py` estimates $\hat{q}$ feed by feed. Persistence is read off the snapshot timeline by rank over the observed poll grid—robust to missed collection polls, which a clock-based rule would miscount as trips: a vehicle whose identifier vanishes for one or more served polls and *reappears* is a linked disappearance, one that vanishes before the window's safe sub-grid and never returns is terminal, and $\hat{q} = \text{linked}/(\text{linked} + \text{terminal})$. The aliasing curve $\kappa(\Delta)$ is obtained by re-reconstructing at coarser intervals, rebalancing β from the operator status flags and departure bursts, and Λ from the reconstructed clean trips. **[To confirm: Tabulate the full per-feed constant table and the cost-feature pipeline (routing, elevation, infrastructure layers) here; software stack and version pins; master seed** $\texttt{SEED} = 42$.**]**

6.2 The compliance–horizon landscape

A first-light audit of 27 French free-floating networks, over a short pilot window with collection ongoing, already exhibits the split the theory predicts. Of the 17 feeds on which q is estimable, 11 *rotate* their identifiers ($\hat{q} \approx 0$: vehicles vanish on rental and never return under the same identifier—e.g. the largest free-floating fleet shows 0 linked against > 2000 terminal disappearances), and 6 *persist* ($\hat{q} \in [0.90, 1.00]$); the remaining 10 are undetermined over the pilot window (low-turnover car-sharing and very small fleets). The distribution is sharply bimodal (Figure 2): $\hat{q}$

[To confirm: Money plot ($T^\star$ vs q, from `d01_pilot.py`): collection horizon against identifier-persistence rate for several trip rates Λ, with the operational-monitoring frontier and the audited feeds positioned on it.]

Figure 1. The compliance–horizon landscape. The required collection horizon $T^\star$ diverges as $q \to 0$ and as δ_{OD} tightens (power -4); feeds below the operational frontier are not monitorable in practice. **[To confirm: regenerate with the multi-day constants.]**

[To confirm: Per-feed identifier-persistence $\hat{q}$ (from `d02_persistence_audit.py`), sorted, with the rotating/persistent thresholds marked.]

Figure 2. Identifier persistence is bimodal across French feeds: $\hat{q}$ is either ≈ 0 (standard-compliant, untrackable) or ≈ 1 (non-compliant, fully trackable), with little mass in between.

clusters at the two extremes, essentially nothing in between, which is exactly the signature of a binary engineering choice—whether or not an operator implements the standard's rotation guidance—rather than a continuum of behaviours. **[To confirm: Replace the pilot-window counts with the multi-day audit once collection matures; report the full constant table and place each network on Figure 1.]** Read through Corollary 6, the rotating half of the panel is simply not OD-monitorable from GBFS at any practical horizon—a conclusion that is about the standard, not the operators, and that we now develop.

7 Discussion: standards and policy implications

Corollary 6 turns a privacy feature of the standard into a quantitative limit on what the standard can be used for. GBFS issue #146 and the v2.0 guidance recommend rotating the vehicle identifier after each trip precisely to prevent the full-resolution OD reconstruction that a persistent identifier enables; the same property, by Theorem 5, is what drives the collection horizon to infinity. There is therefore an irreducible tension, which we make explicit rather than paper over: a feed cannot be simultaneously privacy-compliant (small q) and OD-monitorable at a practical horizon.

For the standard's maintainers (MobilityData) this yields a concrete recommendation. If GBFS is to remain the interface through which agencies expect to derive mobility statistics, it must either (i) specify a minimum, auditable persistence regime—e.g. session-scoped but trip-stable identifiers—or (ii) state plainly that OD is *out of scope* for a privacy-compliant GBFS and route transactional needs to the regulator-facing Mobility Data Specification (MDS), whose two-way, trip-level access is the appropriate channel when $q < q_{\min}$. **[To confirm: Develop the $q_{\min}$ figure for representative French agencies and connect to the LOM/GDPR context.]** Either way, the design choice should be made knowingly, with the horizon law in view, rather than emerging by accident from an identifier-rotation default.

8 Conclusion and limitations

We have shown that the feasibility of OD reconstruction from GBFS is governed by the standard's identifier-persistence regime: the learnable cost is identifiable only up to station effects, its observation bias is exactly the non-separable part of the log-selection (so station emptiness is harmless but 60-second aliasing is not), and the resulting estimator obeys a sampling-horizon law $T^\star \propto \delta_{\mathrm{OD}}^{-4} q^{-1}$ that defines a compliance threshold below which monitoring is infeasible. An audit of French feeds confirms a sharp compliant/non-compliant split.

The results are conditional on the Gibbs/low-rank-cost assumption, which—though it coincides with the field's standard estimators—is the binding commitment; the stability constants in Theorem 5 can be loose at large N (Mena and Niles-Weed, 2019; Genevay et al., 2019); and the estimator recovers the entropy-optimal representative, not pair-level OD, whose interior remains weakly identified. Validation against ground truth requires a feed that both exposes persistent identifiers and publishes trips, a thin intersection we address in future work alongside the multi-day audit and the hierarchical cost of Section 5.1.

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Data availability
All data sources and per-experiment derived outputs are released under the MIT licence at `https://github.com/cycling-data-lab/gbfs-od-reconstruction`. A frozen Zenodo snapshot of the v1.0 release tag is deposited at `https://doi.org/10.5281/zenodo.XXXXXXX` **[To confirm: substitute the issued Zenodo DOI after the first release]**.

Reproducibility
Each table and figure is produced by a numbered script `experiments/d<NN>_<name>.py` that consumes a known input cache, writes a named output JSON/CSV, and pins `SEED=42`. The persistence audit (Section 6) is `d02_persistence_audit.py`; the horizon figure is `d01_pilot.py`. **[To confirm: Add experiment-specific detail.]**

Competing interests
The authors declare no competing financial or non-financial interests.

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